

Graphplay

A Combinatorial Assembly Language for Quantum Primitives

A Typed, Polymorphic, Verified, Categorically-Organized
Combinator Calculus over the Five-Tower Equitable Spine

Anti Mathematics Online Research Crew — Graphplay v3 draft

Abstract

We propose, develop, and partially mechanize **Graphplay**: a typed, polymorphic, verified, categorically-organized combinator calculus for compiling specifications of quantum-walk primitives — perfect state transfer (PST), uniform and instantaneous mixing, spatial search, sampling, state transfer along tails, fractional revival — to host operators on finite weighted graphs, signed/chiral graphs, operator algebras, quantum graphs, and graphons. The core observation is that nearly every published construction in the Tamon corpus and adjacent CTQW literature factors through an **equitable partition** and its small finite quotient matrix. We formalize that factorization as an instruction set: eight load-bearing combinators (PARTITION, QUOTIENT, BUNDLE, SIGN, REFINE, LIFT, COLIMIT, EMBED) which produce, compose, and certify equitable data across a five-tower spine. The compiler signature is

$$(\text{primitive}, \text{hardware}) \mapsto (\text{host}, \text{partition}, \text{schedule}, \text{certificate}),$$

where each output is a Lean 4 term and the certificate is a proof of the relevant spine hypothesis.

Three results are genuinely new with respect to the Tamon corpus. First, the spine lifts cleanly to graphons (Tower 4): we give graphon equitable partitions, prove the spectral-lift theorem for the Hilbert–Schmidt integral operator, and lift PST and spatial-search optimality to the graphon limit. Second, chiral signings are first-class instructions: the characteristic-projection identity `signedBy_preserves_equitable` puts the Levine–Mesapam–Mustico–Tamon–Tucker–Zhan uniform-mixing speedup, and the Xie–Kay–Tamon speed-limit-breaking PST protocol, into one toolkit language — both optimizations factor through a small signed quotient. Third, a categorical filtered-colimit-preservation theorem makes the slogan “quasi-infinite” precise: a filtered family of finite CTQW problems with bounded equitable quotients has its behavior controlled by a single colimit quotient, recovering the Xie–Tamon infinite-tail result and the Bernard–Tamon–Vinet–Xie tails-transfer companion as one-line corollaries.

The Lean 4 development covers Towers 1–5 of the spine, the bundle engine, the chiral specialization, and a graphon scaffold. Companion notes (`research_program.typ`, `applied_majorana1.md`, `applied_ibm_heavy_hex.md`) spin out the dowsing-rod open theorems, cross-disciplinary integrations (Hodge, lattice gauge, MERA, TQFT, WL, RMT, optimal transport, mean-field games), conjectural Towers 6–7, and worked applied case studies; the main paper stays on the spine.

1. Graphplay as an Assembly Language

Thesis. **Graphplay** is a typed, polymorphic, verified, categorically-organized combinator calculus for compiling quantum-primitive specifications to host operators. The “operator” can be a finite Hermitian weighted graph, a chiral signing, an operator-algebra-level quantum graph, or a graphon; the “primitive” is one of PST, uniform mixing, spatial search, sampling, transfer-along-a-tail, fractional revival. The compiler is **typed** because each instruction has a signature on a category of operator-with-partition data; **polymorphic** because the same instruction works across the five towers; **verified** because every instruction has a Lean 4 proof of its specification; **categorically**

organized because the towers are linked by functors and the limit theorems are filtered-colimit preservations.

1.1 The instruction set. The whole stack is built from eight combinators, each polymorphic over a tower-level type:

```

PARTITION : Operator -> (Operator, EquitablePartition)      -- search a
partition
QUOTIENT   : (Operator, EquitablePartition) -> SmallOperator -- shrink to r
x r
BUNDLE     : (Base, Fibers, Couplings) -> Operator          -- build a host
SIGN       : (Operator, Phasing) -> Operator                -- chiral
specialization
REFINE     : (EquitablePartition, Marking) -> EquitablePartition
LIFT       : Primitive(SmallOperator) -> CellUniformPrimitive(Operator)
COLIMIT    : Diagram(Operator, EquitablePartition) -> Operator $\infty$ 
EMBED      : Operator_TowerN -> Operator_TowerN+1

```

A **program** is a sequence of these combinators consumed by a **primitive request** (Section 8). A **certificate** is a Lean term that witnesses the equitability hypothesis each combinator demands.

1.2 Semantics by tower. The eight combinators are not free symbols; each carries a typed semantic clause in each tower. The clauses (proven, modulo flagged open holes, in Lean 4) are:

Combinator	Tower 2 (Weighted)	Tower 3 (Op alg)	Tower 4 (Graphon)
PARTITION	cell-pair row sum const.	$\Pi A \Pi = A \Pi$	a.e.-const cell integrals
QUOTIENT	$r \times r$ Hermitian matrix	bimodule on M_0	same $r \times r$ matrix
BUNDLE	Tot(Q,H,B)	crossed-product analogue	block-step graphon
SIGN	signedBy	unitary $u \in M^*$	a.e.-unimodular kernel
REFINE	marked split	coarser commutant	measurable join
LIFT	Theorem 3.x	Theorem 3.x	Theorem 6.x
COLIMIT	refinement direct limit	direct limit of C^*	cut-norm limit
EMBED	identity	diagonal subalgebra	pixel-step graphon

1.3 The five-tower spine. The towers in order of increasing generality:

Set $\longrightarrow$ SimpleGraph $\longrightarrow$ (Weighted $\cup$ Chiral) $\longrightarrow$ (OpAlg $\cup$ QuantumGraph) $\longrightarrow$ Graphon $\longrightarrow$ Category.

We refer to these as Towers 0–5; the load-bearing combinators are defined and certified at every tower they sensibly inhabit. The Tower-2 through Tower-5 sections (3–7 below) state a **single load-bearing theorem** per tower from which the corner-case results of the literature fall out.

1.4 The compiler signature. The whole stack is consumed by a request

$$\text{Request} = (\text{primitive}, \text{hardware})$$

where the **primitive** is one of PST, UniformMixing, Search, Sample, Transfer, FractionalRevival and the **hardware** is a decidable conjunction of host constraints (planar, surface genus, near-neighbor radius, allowed phase set, vertex degree, size bound). The compiler emits

(host, partition, schedule, certificate)

where **host** is a GraphBundle, **partition** is an equitable partition of the host, **schedule** is a (possibly time-dependent, possibly chiral) Hamiltonian, and **certificate** is a Lean term whose type is exactly the load-bearing-theorem hypothesis for the primitive at the tower in question. The classical Tamon-corpus pipeline (PST on quotient graphs, graph-product PST, spectral spatial-search optimality, tails-transfer, chiral mixing speedup) is the catalog of (primitive, hardware) pairs the compiler can already serve.

1.5 A worked instruction trace. The compilation of “PST between antipodal vertices of a complete multipartite graph $K_{n,n,n,n}$ ” is the trace

```
Request  : (PST, planar + equal-fiber, |V| <= 16)
          |
BUNDLE   : (Q := K_4, H_i := 0, B_(ij) := 1)
          |
PARTITION : pi := { V_1, V_2, V_3, V_4 } (the four color classes)
          |
LIFT      : (PST on K_4 between cell 1 and cell 3) ==> (PST on Tot)
          |
SIGN      : (optional; trivial s = 1)
          |
COLIMIT   : (omitted; finite-stage program)
          |
EMBED     : Tower 2 -> Tower 4 if asked for the graphon limit
```

Every step is type-checked: BUNDLE consumes a Tower-2 base and produces a Tower-2 host; PARTITION consumes a Tower-2 host and produces a Tower-2 host with an EquitablePartition; LIFT consumes a Tower-2 host-with-partition and a small-operator primitive certificate and produces a host-level primitive certificate. The final certificate is the inhabitant of Certificate PST Tot pi schedule.

1.6 Provenance. Two decades of the Tamon program have driven home that the right way to think about CTQW is structural: phenomena that **look** like analytic miracles — PST at integer times [bachman2011quotient], optimal Grover-like search [chakraborty2020spatial, chan2022shadows], uniform mixing at carefully chosen times [levine2026chiral], speed-limit-breaking PST [xie2022speedlimit], optimality on infinite tails [xie2023tail], fractional revival on schemes [chan2019fractional], PST with tails attached [bernard2022tails], PST via products [ge2010products] — are very often projections of small finite calculations to which the equitable partition is the right lens. Graphplay v2 [amorc2025v2] reified part of this picture as a Lean 4 artifact with a GraphBundle engine and a graphon scaffold; v3 changes the lead. We treat the whole stack as an assembly language with the instruction set above and the five-tower spine below as its target. The non-Tamon CTQW papers we build on — spatial search on planar networks [sadowski2014planar], long-range-tunneling searches [king2025longrange], deterministic search on Laplacian-integral graphs [li2025deterministic], lackadaisical search [wong2017lazy], and the surface-topology algebraic-connectivity ceiling [freitas2001heawood] — become (primitive, hardware) instances in the catalog.

1.7 Why this framing is load-bearing. The instruction-set framing pays in three concrete ways. (a) **Polymorphism:** the same PARTITION + LIFT lemma can be used at every tower; the proofs of Theorems 2.2, 3.3, 6.2, 7.1 are structurally the same, varying in the ambient analysis (finite linear algebra, C^* -algebraic projections, L^2 kernel operators, filtered colimit cocones). (b) **Composability:** a verified BUNDLE of two verified BUNDLES is a verified BUNDLE, because the equitability conditions of Theorem 4.3 compose. The compiler can therefore stitch together case-study templates (planar K_4 ,

surface Heawood, IBM heavy-hex, Majorana topological fabric) without re-proof. (c) **Falsifiability**: the conjectural Conjecture 9.3 (Section 10) cleanly links three instructions (COLIMIT, SIGN, embedded QUOTIENT into Bose–Mesner Tower 3) into an iff that is open and supported by worked examples; the assembly-language framing is what makes the conjecture even statable.

2. Tower 2: The Spine and Its Single Load-Bearing Theorem

A **weighted graph** on a finite vertex set V is a Hermitian matrix $A \in \mathbb{C}^{V \times V}$ with zero diagonal. The class includes simple graphs (via 0/1 adjacency), signed and chiral graphs (unitary phases off the diagonal), and weighted spin networks. Lean carrier: `Graphplay/Weighted.lean`.

Definition 2.1 (equitable partition). A partition π of V into nonempty cells $C_1, \dots, C_r$ is **equitable** for A if for every pair of cells C_j, C_k the row sum

$$b_{jk} := \sum_{w \in C_k} A_{vw}$$

is independent of $v \in C_j$. The **quotient matrix** $A/\pi \in \mathbb{C}^{r \times r}$ has $(A/\pi)_{jk} = b_{jk}$. The **characteristic matrix** $P \in \mathbb{C}^{V \times r}$ has $P_{vj} = 1$ if $v \in C_j$ and 0 otherwise. The defining identity is

$$AP = P(A/\pi).$$

This data lives in `Graphplay/Equitable.lean`. The instruction-level reading is that PARTITION produces π and QUOTIENT produces A/π , while EMBED from Tower 1 to Tower 2 is the inclusion of 0/1 matrices into Hermitian matrices.

Theorem 2.2 (single load-bearing theorem of Tower 2). Let π be equitable for A with characteristic matrix P . Then:

1. (Spectrum.) $\text{spec}(A/\pi) \subseteq \text{spec}(A)$; for every eigenvector x of A/π with eigenvalue λ , Px is a cell-constant eigenvector of A with eigenvalue λ .
2. (CTQW restriction.) e^{-itA} commutes with $\Pi := P(P^*P)^{-1}P^*$ and acts on range Π as $e^{-itA/\pi}$.
3. (Primitive lift.) For any cell-uniform primitive request $p \in \{\text{PST, UniformMixing, Search, Transfer, Sample, FractionalRevival}\}$ whose endpoints (and marked sets, for Search) are unions of cells, p holds on (A, π) iff p holds on A/π after the canonical refinement.

The three clauses together are the LIFT instruction at Tower 2. Clause (i) is the classical Godsil spectral lift; clause (ii) is the CTQW invariance lemma; clause (iii) packages [bachman2011quotient, ide2022equitable, chan2022shadows] as a single combinator. Lean: `Graphplay/Spectral.lean`, `Graphplay/PST.lean`, `Graphplay/Search.lean`, `Graphplay/Mixing.lean`.

Proof sketch. For (i): suppose $(A/\pi)x = \lambda x$. Then $APx = P(A/\pi)x = \lambda Px$. Since P has full column rank (because the cells are disjoint and nonempty), $Px \neq 0$ for $x \neq 0$, so $\lambda \in \text{spec}(A)$ with eigenvector Px . For (ii): the identity $AP = P(A/\pi)$ implies $A\Pi = P(P^*P)^{-1}(A/\pi)P^* = \Pi A\Pi$ on the range of Π , so A commutes with Π (using Hermiticity); the unitary e^{-itA} then commutes with Π and acts on range Π as the unitary $e^{-it\Pi A \Pi} = e^{-itA/\pi}$ identified via the cell-uniform basis. For (iii): the three primitive functionals $|\langle \psi_t, e_a \rangle|^2$ (PST), $1/N^2 \sum |U_{t(v,w)}|^2$ (uniform mixing on cells), $|\langle e_w, e^{-itH_\gamma} e_a \rangle|^2$ (search) are each polynomial in matrix entries that lie in range Π when initial and final states do, hence equal to the same polynomial in the quotient. $\square$

The marked-vertex case (relevant to Search) is worth its own statement:

Corollary 2.3 (search refinement). For a marked vertex $w \in V$, the refinement π_w that splits the cell containing w into $\{w\}$ and its complement within that cell is equitable for the search Hamiltonian

$H_\gamma := -\gamma A - e_w e_w^*$. The unitary e^{-itH_γ} restricts to the $(r+1)$ -dimensional cell-uniform-plus- $\{w\}$ subspace. The optimality criterion of [chakraborty2020spatial], sharpened by [chan2022shadows], reduces to a spectral computation on this small subspace. The marked-cell refinement idiom is due to [ide2022equitable].

Examples.

1. (Complete multipartite.) $K_{n_1, \dots, n_r}$ has the color partition as equitable, with $(A/\pi)_{jk} = n_k$ for $j \neq k$, otherwise 0.
2. (Xie–Tamon tail [xie2023tail].) $K_n + P_n$ has the 2-cell partition (clique, tail) as equitable; the quotient is a 2×2 Hermitian matrix encoding Farhi–Gutmann complete-graph search.
3. (BTV-Xie tails-PST [bernard2022tails].) The transfer-side companion: PST between two distinguished vertices on a finite block with arbitrarily long tails is equivalent to PST on a 2-cell quotient.
4. (Heavy-hex search.) The IBM heavy-hex lattice, when colored so that two cells coincide with the marked-vertex pair, has a 4×4 quotient whose spectrum is computed in companion `applied_ibm_heavy_hex.md`.
5. (Topological search [freitas2001heawood, sadowski2014planar].) On a surface of genus g , the algebraic connectivity is bounded above by the Heawood number $H(g)$; via the bundle engine of Section 4 this is a Tower-2 budget on the template Q .

3. Tower 3: Operator Algebras, Coherent Algebras, Quantum Graphs

The next tower sits above weighted graphs. Let $\mathcal{A} \subseteq M_{V(\mathbb{C})}$ be the unital $*$ -subalgebra generated by A and the identity. An equitable partition π corresponds to the orthogonal projection $\Pi = P(P^*P)^{-1}P^*$ onto the cell-constant subspace; the equitability condition is exactly $\Pi A \Pi = A \Pi$, i.e.

Π commutes with A . Lean: `Graphplay/QuantumGraph.lean`.

Definition 3.1 (quantum graph). A **quantum graph** on a finite-dimensional C^* -algebra $M \subseteq B(H)$ is a self-adjoint M -bimodule $S \subseteq B(H)$ with adjacency operator A_S . An **equitable coarsening** is a unital $*$ -subalgebra $M_0 \subseteq M$ such that the orthogonal projection $\Pi_{\{M_0\}}$ onto the M_0 -invariant vectors commutes with A_S .

Definition 3.2 (partition-projector algebra). The **partition-projector algebra** $\mathcal{P}(\pi) \subseteq M_{V(\mathbb{C})}$ is the subalgebra of cell-pair-constant matrices. This is the maximal subalgebra in which π is equitable; it is closed under Schur and matrix products and is the smallest coherent algebra containing Π .

Theorem 3.3 (single load-bearing theorem of Tower 3). Let S be a quantum graph on M with adjacency A_S , and let M_0 be an equitable coarsening with projection Π . Then:

1. (Spectrum.) $\text{spec}(\Pi A_S \Pi) \subseteq \text{spec}(A_S)$.
2. (CTQW restriction.) e^{-itA_S} commutes with Π and acts on range Π as $e^{-it\Pi A_S \Pi}$, the adjacency of a smaller quantum graph S/M_0 on M_0 .
3. (Primitive lift.) Cell-uniform PST, UniformMixing, Search, Transfer, and FractionalRevival lift from S/M_0 to S .

The Tower-2 spine of Theorem 2.2 is the special case $M_0 := \text{image of the diagonal subalgebra under coarsening by } \pi$. Lean: `Graphplay/QuantumGraph.lean`.

Proofsketch. For (i): the operator A_S commutes with Π by assumption, so A_S and Π are simultaneously diagonalizable; the spectrum of $\Pi A_S \Pi$ is the restriction of $\text{spec}(A_S)$ to range Π . For (ii): e^{-itA_S} is a strong limit of polynomials in A_S , each of which commutes with Π . For (iii): on the bimodule S , the cell-uniform-plus-marked subspace of the search refinement remains a left and right M_0 -module,

so the search dynamics close on V_π with an effective adjacency $\Pi A_S \Pi - \gamma \Pi P_S \Pi$. The primitive functionals are again polynomial in the restricted operator. $\square$

Remark 3.4 (commutative versus non-commutative). In the commutative case ($M = \text{diag}$), Tower 3 reduces to Tower 2 with no new content. The Tower-3 generality shows itself in the non-commutative case: a quantum graph whose bimodule is non-trivially multidimensional admits equitable coarsenings only when the bimodule respects $\Pi_{\{M_0\}}$, a strictly stronger condition than the diagonal equitable partition. Cf.

Graphplay/Dowsing/NonCommutativeCoherent.lean.

Examples.

1. (Bose–Mesner.) The Bose–Mesner algebra of a commutative association scheme is the maximally symmetric Tower-3 coarsening. Fractional revival on Hamming schemes [chan2019fractional] is the Tower-3 corollary of a Tower-2 search-lift on the scheme partition.
2. (Coherent algebras.) The Schur-and-matrix-product closure of $\mathcal{P}(\pi)$ is the coherent algebra of π ; this is the right context for the Weisfeiler–Leman refinement chain.
3. (Quantum-graph reduction.) A non-commutative quantum graph admits an equitable coarsening iff its bimodule respects the projection $\Pi_{\{M_0\}}$; this is the operator-algebra version of the search-lift idiom of [ide2022equitable].

4. The Constructive Engine: ‘BUNDLE’ and its Corners

The Tower-2 / Tower-3 spine tells us what is needed; the BUNDLE instruction provides hosts that satisfy it constructively. The point of the construction is that products, joins, color completions, and engineered templates are all corners of one definition.

Definition 4.1 (graph bundle). A **graph bundle** $\text{GraphBundle } Q \ V$ is the data

$$Q : \text{WeightedGraph}(I), \quad V : \forall i \in I, V_i, \quad H_i : \text{WeightedGraph}(V_i), \quad B_{ij} : V_i \times V_j \rightarrow \mathbb{C},$$

subject to a Hermitian compatibility $B_{ij}(a, b) = \overline{B_{ji}(b, a)}$ only required where $Q_{ij} \neq 0$. The **total graph** $\text{Tot}(Q, H, B)$ on $\coprod_i V_i$ has

$$\text{Tot}(x, y) = \begin{cases} H_{i(x,y)} & \text{if } x, y \text{ in same fiber } V_i \\ Q_{ij} \cdot B_{ij}(x, y) & \text{if } x \in V_i, y \in V_j, i \neq j \end{cases}.$$

Lean: Graphplay/Bundle.lean.

Proposition 4.2 (corners).

1. $H_i = 0, B_{ij} = 1$ on Q -edges: the $\text{TemplateJoin}(Q, V)$ of v2.
2. $Q = K_2, H_1, H_2$ arbitrary, $B_{12} = 1$ on a chosen product subset: Cartesian, lexicographic, strong, tensor, conormal, and disjunctive products as cases. The PST-product theorems of [ge2010products] fall out as Tower-2 corollaries.
3. $Q = K_C$ on a color palette, $H_i = 0$: color completion; with $B_{ij} = 1$, complete multipartite.
4. Q a join graph in the sense of [kirkland2023join], arbitrary fibers: the Kirkland–Monterde join construction; their PST-on-joins characterization is a Tower-2 search-lift statement.
5. Q a Heawood-template surface graph: the surface completion pipeline of v2, with a Tower-2 spectral budget from [freitas2001heawood].

Theorem 4.3 (BUNDLE is equitable iff regular fibers + biregular couplings). The cell partition $\pi := \{V_i\}_{i \in I}$ is equitable for $\text{Tot}(Q, H, B)$ if and only if

1. each fiber H_i is regular with constant row sum r_i , and

2. each coupling B_{ij} is biregular: $\sum_{b \in V_j} B_{ij}(a, b)$ is independent of a , and $\sum_{a \in V_i} B_{ij}(a, b)$ is independent of b .

When these conditions hold, $(A/\pi)_{ij} = r_i \delta_{ij} + Q_{ij} |V_j| \beta_{ij}$ where β_{ij} is the constant row sum of B_{ij} . Lean: `Bundle.equitable_iff_regular_biregular`.

Corollary 4.4 (spectral inheritance). For equal-fiber bundles ($|V_i| = m$ for all i , $H_i = 0$, $B_{ij} = 1$):

$$\text{spec}(\text{Tot}(Q, H, B)) = m \cdot \text{spec}(Q) \cup (\text{internal zero modes}).$$

The CNO criterion [chakraborty2020spatial] in the [chan2022shadows]-sharpened form, the deterministic-search criterion of [li2025deterministic], and the long-range-tunneling regime of [king2025longrange] are inherited by the bundle from the template.

The engineering reading is that the compiler’s **certificate** for a bundle host is exactly the pair (regular fibers, biregular couplings) of Theorem 4.3 — both decidable, both Lean-checkable.

Proof sketch of Theorem 4.3. ($\Leftarrow$.) Fix cells V_i and V_j . For $x \in V_i$, the row sum of $\text{Tot}(Q, H, B)$ from x to V_j is $H_{i(x, V_j)} \cdot \delta_{ij} + Q_{ij} \sum_{b \in V_j} B_{ij}(x, b)$, which equals $r_i \delta_{ij} + Q_{ij} |V_j| \beta_{ij}$ by the regularity-plus-biregularity hypothesis — independent of x in V_i . ($\Rightarrow$.) If row-sums into V_j depend only on the cell of x , specialize to $j = i$ to get fiber regularity; specialize to $j \neq i$ with $Q_{ij} \neq 0$ to get the biregularity of B_{ij} on the first slot; the dual constraint (independence of y in V_j) requires biregularity on the second slot. $\square$

Example 4.5 (Heawood completion at genus g). Take $Q = K_{H(g)}$, where $H(g)$ is the Heawood number of the surface, fibers $H_i = 0$, couplings $B_{ij} = 1$. Theorem 4.3 is satisfied. The quotient is $A/\pi = (H(g) - 1)(J - I)$ on $H(g)$ vertices, whose spectrum consists of $H(g) - 1$ with multiplicity 1 and -1 with multiplicity $H(g) - 1$. The Tower-2 spectral budget of [freitas2001heawood] is the statement that the spectral gap of the host is $H(g)$, capturing the surface-topology obstruction.

Example 4.6 (rook graph $K_3 \square K_3$). The Cartesian product $K_3 \square K_3$ is the bundle with $Q = K_3$, fibers $H_i = K_3$, couplings $B_{ij} = I$. Theorem 4.3 is satisfied; the quotient is a 3×3 matrix with diagonal 2 and off-diagonal 1, giving spectrum $\{4, 1, 1\}$, which combined with internal modes recovers the full rook spectrum $4, 1, 1, 1, 1, -2, -2, -2, -2$. This is the `v2 rook_3x3_equal_fiber` example.

Example 4.7 (long-range tunneling [king2025longrange]). Take $Q = K_n$, fibers $H_i = 0$, couplings $B_{ij}(x, y) = \exp(-\alpha d(x, y))$ for a power-law decay rate α and an underlying metric d on each fiber identified across fibers. The biregularity condition of Theorem 4.3 is satisfied iff $\sum_{y \in V_j} e^{-\alpha d(x, y)}$ is independent of x , which holds when the metric is vertex-transitive on each fiber. The deterministic-search criterion of [li2025deterministic] on the resulting Laplacian-integral host is then a Tower-2 spectral diagnostic on Q .

5. Tower 2 Continued: ‘SIGN’ and the Chiral Specialization

The chiral story — unitary phases on edges of a weighted graph — has been known since Sedlacek to permit phenomena unavailable to real weights. Two recent works exhibit this loudly: [levine2026chiral] gives chiral signings that produce uniform mixing on K_n^σ and identifies an arithmetic obstruction governing the phenomenon; [xie2022speedlimit] gives a dual-rail chiral protocol on an engineered chain that **breaks the perfect-state-transfer speed limit**. Both results are quotient phenomena in Graphplay.

Definition 5.1 (chiral signing). A **chiral signing** is a function $s : V \rightarrow U(1)$; the **signed graph** G^s has adjacency $G_{vw}^s = \overline{s(v)} G_{vw} s(w)$. Lean: `Graphplay/Chiral.lean`. The signing is **cross-constant** with respect to a partition π if $s(v)s(w)$ depends only on the pair of cells of v, w .

Theorem 5.2 (SIGN preserves PARTITION, signedBy_preserves_equitable, proven). If π is equitable for G and s is cross-constant with respect to π , then π is equitable for G^s , and the signed quotient satisfies

$$G^s/\pi = (G/\pi)^{s_\pi},$$

where s_π is the inherited cross-constant phase on cell-pairs. In particular, CTQW on G^s restricts to the cell-constant subspace and acts there as $e^{-it(G/\pi)^{s_\pi}}$. Lean: `WeightedGraph.signedBy_preserves_equitable` in `Graphplay/Chiral.lean`, fully proven.

Theorem 5.3 (SIGN optimization factors through the quotient). For a bundle $\text{Tot}(Q, H, B)$ with cross-constant signing and phase-equitable cell partition π , the uniform-mixing functional

$$M(s; t) := \max_{v, w} |U_{s, t}(v, w)|^2$$

on cell-uniform initial and final states equals the corresponding functional for the signed quotient $(G/\pi)^{s_\pi}$. Optimizing the signing over the bundle phase space therefore reduces to optimizing s_π over the small phase space $U(1)^{E(Q)}$, modulo gauge. Lean: `Graphplay.Chiral.chiral_mixing_optimization`.

Corollary 5.4 (Levine–Mesapam–Mustico–Tamon–Tucker–Zhan [levine2026chiral] as a quotient corner). The uniform-mixing speedup on K_n^σ has a bundle version: if a bundle has K_n^σ as its quotient and equal fibers, the corresponding chiral mixing speedup holds for the bundle on the cell-uniform subspace. The witness K_4 signing reproducing $K_1 + K_{1,3}$ is `Chiral.unitaryHammingChiralK4Signing` in Lean.

Corollary 5.5 (Xie–Kay–Tamon [xie2022speedlimit] as a quotient corner). The dual-rail chiral protocol of **Breaking the speed limit for perfect quantum state transfer** is naturally a 2×2 chiral quotient with an off-diagonal phase implementing the anti-Zeno-effect timing. Concretely: an engineered spin chain with the Xie–Kay–Tamon weights and dual-rail readout admits an equitable partition whose quotient is exactly the 2×2 signed Hermitian matrix studied in their protocol, and the speed-limit-breaking transfer time is a function of that small matrix. The toolkit catalog entry is “given target transfer time τ below the spectral speed limit, request (host, partition, schedule, certificate)”. A tight characterization of which engineered hosts permit a target τ as a function of the chiral quotient norm remains open.

The Lean development covers all of Section 5 except a tight version of Corollary 5.5, which is flagged in `Graphplay/Dowsing/ChiralBundlePST.lean` as `OpenQ3_chiralHeawoodPGST` and three adjacent open questions.

Proof sketch of Theorem 5.2. Equitability of π for G^s requires that the cell-pair row sum $\sum_{w \in C_k} \overline{s(v)} G_{vw} s(w)$ depend only on the pair (C_j, C_k) for $v \in C_j$. Factor out $\overline{s(v)}$: the sum becomes $\overline{s(v)} \sum_{w \in C_k} G_{vw} s(w)$. The cross-constancy hypothesis says $s(w) = c_{jk} s(v)$ for some unit c_{jk} when $w \in C_k, v \in C_j$, so $\sum_{w \in C_k} G_{vw} s(w) = c_{jk} s(v) \sum_{w \in C_k} G_{vw} = c_{jk} s(v) b_{jk}$ by the original equitability. Therefore $\overline{s(v)} \sum_{w \in C_k} G_{vw}^s = c_{jk} b_{jk}$, depending only on the pair. The signed quotient identity follows directly. $\square$

Remark 5.6 (cross-constant versus phase-equitable). The hypothesis of cross-constancy is strictly stronger than phase-equitability (= signed row-sum constancy). Cross-constancy means the phase is **gauge-equivalent** to a Tower-2 chiral signing of the small quotient; phase-equitability allows fiber-internal phases that cell-uniform observables never see. Corollary 5.4 is in the cross-constant regime; Corollary 5.5 may not be, and that is exactly what `OpenQ3_chiralHeawoodPGST` flags.

Example 5.7 (Levine signing on K_4). The witness K_4 signing whose quotient mimics $K_1 + K_{1,3}$ is $s(0) = 1, s(1) = i, s(2) = -1, s(3) = -i$, giving signed adjacency

$$G^s = \begin{pmatrix} 0 & i & -1 & -i \\ -i & 0 & i & -1 \\ -1 & -i & 0 & i \\ i & -1 & -i & 0 \end{pmatrix}.$$

This signing is cross-constant for the trivial single-cell partition; the resulting uniform-mixing time on K_4^s is strictly shorter than the real K_4 mixing time. This is the smallest instance of [levine2026chiral] and is the seed for `Chiral.unitaryHammingChiralK4Signing`.

6. Tower 4: Graphons and the Quasi-Infinite Limit Theorem (★)

We now lift the spine to graphon kernels. This is the headline novel section.

A **Hermitian graphon** on a probability space (Ω, μ) is a measurable function $W : \Omega \times \Omega \rightarrow \mathbb{C}$ with $W(x, y) = \overline{W(y, x)}$ and $W \in L^2$. The induced integral operator $T_W : L^2(\mu) \rightarrow L^2(\mu)$, $(T_W f)(x) = \int W(x, y) f(y) \, d\mu(y)$, is compact, self-adjoint, and Hilbert–Schmidt. The CTQW on W is the unitary group $U_{W(t)} := e^{-itT_W}$. Lean carriers: `Graphplay/Graphon.lean`, `Graphplay/Graphon/Equitable.lean`, `Graphplay/Graphon/PST.lean`, `Graphplay/Graphon/Limit.lean`.

Definition 6.1 (graphon equitable partition). A measurable partition $\Omega = \bigsqcup_{k=1}^r A_k$ is **equitable** for W if for all j, k the function

$$x \mapsto \int_{A_k} W(x, y) \, d\mu(y)$$

is constant almost everywhere on A_j , with value $\tilde{b}_{jk}$. In the rescaled-indicator basis $u_k := \mu(A_k)^{-1/2} \mathbb{1}_{A_k}$, the **graphon quotient matrix** $W/\pi \in \mathbb{C}^{r \times r}$ has entries

$$(W/\pi)_{jk} = \tilde{b}_{jk} \sqrt{\mu(A_k)/\mu(A_j)}.$$

The closest finite ancestor is the Bick–Sclosa invariant-subspace decomposition [bick2024graphlimits] for **real** dynamics on graphons; we record the operator-theoretic **unitary** version. The operator-theoretic LQR background is Gao–Caines [gao2020graphon]; the spectral graph-limit theorems are Szegedy [szegedy2010spectral].

Theorem 6.2 (★ single load-bearing theorem of Tower 4). For a Hermitian graphon W with measurable equitable partition π of finite size r :

1. (Spectrum.) $\text{spec}(W/\pi) \subseteq \text{spec}(T_W)$; eigenvectors x of W/π lift to eigenfunctions $\sum_k x_k u_k$ of T_W with the same eigenvalue.
2. (CTQW restriction.) $U_{W(t)}$ preserves the finite-dimensional cell-constant subspace $V_\pi := \text{span}\{u_k\}$ and acts there as $e^{-itW/\pi}$.
3. (Primitive lift.) If perfect state transfer between two cell-uniform states occurs in the quotient at time τ , it occurs in the graphon walk between the corresponding cell-uniform states of $L^2(\mu)$. For a measurable marked set $S \subseteq \Omega$ of positive measure, the search Hamiltonian $H_{W, \gamma, S} := -\gamma T_W - \text{Proj}_{L^2(S)}$ has the refined partition $\pi_S := \pi \bowtie \{S, S^c\}$ equitable, and the search dynamics restrict to a Hamiltonian of dimension $\leq 2r$ on V_{π_S} .

This is a graphon-level generalization of [bachman2011quotient] and, to our knowledge, has no prior published analogue at this level of unification: the closest is the **real-dynamics** invariant-subspace lift of [bick2024graphlimits]. We state and partially mechanize it in `Graphplay/Graphon/PST.lean`.

Proof sketch. (i) Apply T_W to u_k : $(T_W u_k)(x) = \mu(A_k)^{-1/2} \int_{A_k} W(x, y) \, d\mu(y) = \mu(A_k)^{-1/2} \tilde{b}_{jk}$ for $x \in A_j$ almost everywhere. Re-expressing in the basis u_j via $u_{j(x)} = \mu(A_j)^{-1/2} \mathbb{1}_{A_j}(x)$ gives $T_W u_k =$

$\sum_j (\tilde{b}_{jk} \mu(A_j)^{1/2} \mu(A_k)^{-1/2}) u_j = \sum_j (W/\pi)_{jk} u_j$ after symmetrizing the normalization under the Hermitian assumption. (ii) This shows T_W maps V_π into itself with matrix W/π ; the unitary $U_{W(t)} = e^{-itT_W}$ inherits the invariance and acts on V_π as $e^{-itW/\pi}$ by spectral calculus. (iii) Cell-uniform PST and search functionals are again polynomial in T_W between cell-uniform states; the polynomial values agree with the finite-matrix calculation. $\square$

Theorem 6.3 (quasi-infinite limit theorem). Let $(G_n, \pi_n)_{n \in \mathbb{N}}$ be a sequence of finite Hermitian weighted graphs with equitable partitions, all of size at most r . Suppose:

1. The pixel-step graphons W_{G_n} converge to a Hermitian graphon W in cut norm.
2. The cell-fraction vectors converge to a probability vector $(\mu(A_1), \dots, \mu(A_r)) \in \Delta^{r-1}$ with all $\mu(A_k) > 0$.
3. The normalized quotient matrices G_n/π_n converge to a Hermitian matrix $M \in \mathbb{C}^{r \times r}$ in Frobenius norm.

Then W admits a measurable equitable partition π realizing the limit cell fractions and with $W/\pi = M$. Moreover, every cell-uniform PST or spatial-search problem on (G_n, π_n) has a well-defined limit on (W, π) , and the limit's value is the value of the finite Hermitian problem on M . Lean: Graphplay/Graphon/Limit.lean; the proof reduces to (a) cut-norm-implies-operator-norm convergence under bounded entries [szegedy2010spectral], (b) Frobenius convergence of the quotient matrices, and (c) continuity of finite spectral optimization problems.

Proof sketch. By [szegedy2010spectral], cut-norm convergence with bounded entries implies operator-norm convergence of $T_{W_{G_n}} \rightarrow T_W$. The indicator functions $\mathbb{1}_{A_{n,k}}$ of the cells of π_n , viewed in L^2 , converge to $\mathbb{1}_{A_k}$ for a measurable limit partition because the cell-fraction vector converges and the measures are uniformly bounded. Apply $T_{W_{G_n}}$ to $\mathbb{1}_{A_{n,k}}$ and use both convergences: in the limit, $T_W \mathbb{1}_{A_k} = \sum_j \tilde{b}_{jk} \mathbb{1}_{A_j}$ a.e., which is the graphon equitability. The values $\tilde{b}_{jk}$ are the limit Frobenius entries. The PST/search functionals are continuous in the finite quotient and in the (uniformly bounded) cell fractions, hence continuous through the limit. $\square$

Corollary 6.4 (Xie–Tamon [xie2023tail] as a 2-cell corner). The sequence $K_n + P_n$ with two-cell partition (clique, tail) satisfies the hypotheses of Theorem 6.3 with limit a complete graphon on $[0, 1/2]$ together with an infinite path of mass $1/2$; the limit quotient is the 2×2 Hermitian matrix encoding the Farhi–Gutmann complete-graph search, and optimality is preserved.

Corollary 6.5 (BTV–Xie [bernard2022tails] as a 2-cell corner). The transfer-side companion is recovered by the same 2-cell scheme applied to the PST primitive: $K_n + P_n$ -like sequences with PST endpoints in the clique block have the limit governed by the same 2×2 matrix, and tails-PST is preserved.

Example 6.6 (cycle blowup). Fix a template weighted graph Q on r vertices. The sequence $\text{Tot}(Q, 0, J)$ with equal fibers of size n , viewed as pixel-step graphons, converges in cut norm to the block-constant graphon W_Q with cells of measure $1/r$ and value Q_{jk} in block (j, k) . The quotient at every stage is $n \cdot Q$ (up to a renormalization absorbed into the time parameter), and Theorem 6.3 gives the limit as the finite search problem on Q . The compiler examples of v2 (rook_3x3_equal_fiber, cycle8_powerlaw_alpha1, torus_heawood7_equal_fiber) are the finite stages of such a sequence.

7. Tower 5: ‘COLIMIT’ as a Filtered-Colimit-Preserving Functor

Let EqWGr be the category whose objects are pairs (G, π) with G a finite Hermitian weighted graph and π an equitable partition, and whose morphisms $(G, \pi) \rightarrow (G', \pi')$ are pairs (φ, θ) with φ a weight-preserving cell-block map and θ refining π to a partition compatible with φ . Let $\text{Mat}_{\text{fin}}(\mathbb{C})$ be the category of finite Hermitian matrices. Define $\text{Quot} : \text{EqWGr} \rightarrow \text{Mat}_{\text{fin}}(\mathbb{C})$ by $\text{Quot}(G, \pi) = G/\pi$.

Theorem 7.1 (single load-bearing theorem of Tower 5). The functor `Quot` preserves filtered colimits. Explicitly, given a filtered diagram (G_α, π_α) in `EqWGr` whose underlying graphs form a sequence with a graphon colimit W (in the cut metric) and whose quotient matrices have a finite-dimensional Frobenius limit M , the colimit pair (W, π) exists in the extended graphon-equitable category and $W/\pi = M$. Lean: `Graphplay/Categorical.lean`, where the chain colimit, cochain limit, and `Quotient.preservesFilteredColimits` are mechanized.

Corollary 7.2 (unification). The quasi-infinite limit theorem (Theorem 6.3) is the Tower-4 instance of Theorem 7.1. The Tower-1 inverse-limit and edge-union universal properties of `v2` are the Tower-1 and Tower-0 instances. The “failed reflector” observation — there is no left adjoint to “forget the coloring” — is the statement that the underlying-graph functor does **not** have a colimit-preserving left adjoint; the **relative** left adjoint along a chosen template Q does exist and is the bundle construction of Section 4 (`Bundle.ofTemplateJoin_total_eq_templateJoin`).

Corollary 7.3 (Xie–Tamon and BTV-Xie via the categorical machine). Theorem 7.1 instantiated on the 2-cell schemes of Corollaries 6.4–6.5 returns Xie–Tamon [xie2023tail] and BTV-Xie [bernard2022tails] as direct images of the same `COLIMIT` instruction.

Proof sketch of Theorem 7.1. Filtered colimits in `EqWGr` are built from refinement-compatible cocones. Given a filtered diagram, the underlying cell-fraction vectors live in Δ^{r-1} (compact); pass to a convergent subnet to extract a limit fraction vector. The quotient matrices live in the bounded subset $\{M : |M|_F \leq C\}$ of $\mathbb{C}^{r \times r}$; extract a Frobenius subnet limit M . The graphs themselves, viewed as pixel-step graphons, live in the cut-metric-compact subset $\{W : |W|_\infty \leq 1\}$ (Lovasz); extract a cut-norm subnet limit W . The three sub-limits assemble into a single colimit object in the extended graphon-equitable category by Theorem 6.3, and the universal property follows from the fact that all cocone maps respect the characteristic matrix. The Lean proof uses Mathlib’s `IsFiltered` API on the index category and the abstract colimit-preservation criterion via representability. $\square$

Worked Tower-5 chain. Take the chain $G_n := K_n + P_n$ with 2-cell partition π_n . The quotient matrices are

$$G_n/\pi_n = \begin{pmatrix} (n-1) + 0 & 1 \\ 1 & 2 + 0 \end{pmatrix}.$$

(Off-diagonal 1 because only one edge connects clique to tail; diagonal $n - 1$ is the clique-internal row sum; diagonal 2 is the path-internal row sum at an interior vertex, with corrections at the endpoints absorbed into the normalization.) Renormalizing so that the cell measure is fixed and the largest entry is bounded, the Frobenius limit as $n \rightarrow \infty$ is a bounded 2×2 Hermitian matrix. The cut-norm limit of the pixel-step graphons of G_n is a Hermitian graphon equal to 1 on $[0, 1/2]^2$, equal to a path-kernel on $[1/2, 1]^2$, and equal to a single bridging edge of measure zero. Theorem 6.3 / Theorem 7.1 gives the limit search functional explicitly, recovering the [xie2023tail] result. The same chain with PST endpoints in the clique block gives the [bernard2022tails] result by Corollary 6.5.

8. Engineering Toolkit: `(primitive, hardware)` to `(host, partition, schedule, certificate)`

We give the compiler its signature.

A **primitive** is a constructor in the Lean enum

`Primitive := PST | UniformMixing | FractionalRevival | Search | Transfer | Sample.`

A **hardware specification** is a decidable predicate on weighted graphs. Graphplay/Toolkit/Hardware.lean provides constants `planarSpec`, `surfaceSpec( $g$ )`, `nearestNeighborSpec( $r$ )`, `chiralAllowedSpec( $\Phi$ )`, `regularSpec( $d$ )`, `sizeBoundedSpec( $n$ )`, their finite intersections, and a **quotient** operation that lifts a hardware specification on the host to a specification on the quotient.

A **schedule** is a piecewise-constant or piecewise-analytic family of Hamiltonians on the host. Graphplay/Toolkit/Scheduler.lean provides static, magnetic-flux, and adiabatic schedules together with `Schedule.evolve_preserves_cellUniform`: cell-uniform states stay cell-uniform under any schedule whose generators preserve the partition projector. This is the certificate that makes the spine usable **time-dependently**.

A **noise model** is a Lindblad-type perturbation; Graphplay/Toolkit/Noise.lean defines a `NoiseModel`, the cell-uniform-symmetric subclass, and `Noise.cellUniform_preserved`, which says the cell-uniform subspace remains invariant under a cell-uniform-symmetric noise model. This lemma promotes Theorems 2.2, 3.3, 5.2, 6.2, 7.1 to noisy-CTQW versions; full quantitative treatment is open.

A **certificate** of a (primitive, host, partition, schedule) tuple is a Lean term of type `Certificate primitive host partition schedule`, with field constructors corresponding to Theorems 4.3, 5.2, and 6.2 as appropriate. The compiler emits both a JSON artifact and a Lean term; the Lean term is checked by lake build before deployment, giving a machine-verifiable bridge from a hardware-and-primitive specification to a quantum-walk schedule. The case studies of v2 are precisely equal-fiber bundles whose templates carry verifiable regularity certificates; they become end-to-end Lean-checked instances of the compiler.

The Certificate type. Schematically,

```
structure Certificate
  (p : Primitive) (B : GraphBundle) (pi : EquitablePartition B.tot)
  (sched : Schedule B) where
  regular_fibers      : ∀ i, Regular (B.H i) (B.fiberRowSum i)
  biregular_couplings : ∀ i j, Q.adj i j ≠ 0 → Biregular (B.B i j)
  schedule_equitable  : ∀ t, EquitablePartition (sched.evolve t).gen pi
  primitive_quotient  : QuotientCertifies p (B.quotient pi)
```

The four fields are: the two regularity certificates of Theorem 4.3, the schedule-side equitable-preservation certificate of `Schedule.evolve_preserves_cellUniform`, and the small-quotient primitive certificate (a finite eigen-decomposition or PST-time witness). All four are decidable in the finite case; the graphon case adds a measurability side-condition. The toolkit ships with constructors for the catalog entries below, each returning a `Certificate` whose Lean term is the artifact deployed to hardware.

The toolkit catalog at v3 contains the following (primitive, hardware) entries with full certificates:

Primitive	Hardware	Host	Reference
PST	near-neighbor chain	Christandl-Datta path	classical
PST	planar, equal-fiber	<code>TemplateJoin(K_4, V)</code>	v2
PST	surface genus g	<code>TemplateJoin($K_{(H(g))}$, V)</code>	v2
PST	dual-rail, chiral	Xie–Kay–Tamon chain	xie2022speedlimit
PST	tail attached	$K_n + P_n$	bernard2022tails
Search	Laplacian-integral	<code>Tot(Q, $\emptyset$, J)</code>	li2025deterministic

Search	long-range tunneling	Tot(Q, 0, B(alpha))	king2025longrange
Search	planar near-neighbor	Tot(K_4, 0, J)	sadowski2014planar
Search	infinite tail probe	graphon limit of K_n + P_n	xie2023tail
UniformMixing	chiral K_n^σ quotient	bundle on K_n^σ	levine2026chiral
FractionalRevival	Hamming scheme	Hamming-scheme bundle	chan2019fractional

9. Spectral Disassembly: Decompiling Operators to Equitable Programs

The compiler so far goes forward: from a (primitive, hardware) request to a (host, partition, schedule, certificate) quadruple. The reverse problem — given a host operator, recover a (smallest) BUNDLE +

PARTITION program that produces it — is **decompilation**. The Lean artifact contains a stub `Graphplay/Toolkit/Spec.lean` for the disassembler and a stub partition-search routine; the analysis is brief here because two companion files do the worked applied case studies:

1. `paper/applied_majorana1.md` — decompilation of the Microsoft Majorana 1 chip Hamiltonian to an equitable program: a topological-qubit fabric with a small chiral template quotient, exhibiting the chiral speed-limit-breaking corner (Corollary 5.5) on engineered chains. The applied question is which Majorana-coupling patterns produce a chiral quotient with a non-trivial speedup window.
2. `paper/applied_ibm_heavy_hex.md` — decompilation of the IBM heavy-hex lattice Hamiltonian: a planar near-neighbor host whose color partition has a 4×4 quotient, with PST and search corners visible at the Tower-2 spine. The applied question is the optimal magnetic-flux phasing consistent with the heavy-hex geometry, framed as a quotient-level optimization (Theorem 5.3).

The main paper records the framework; the case studies live in those companion notes.

9.1 The disassembler signature. Decompilation is the search problem

$$\text{Disasm} : \text{Operator} \rightarrow \text{set} (\text{BundleProgram} \cup \text{Failure}),$$

where a `BundleProgram` is a sequence of PARTITION, BUNDLE, SIGN, and REFINE instructions producing the input operator. The two canonical algorithms are

1. **Weisfeiler–Leman refinement** as a PARTITION-search engine. Starting from the trivial partition, iteratively refine by row-sum signature until stable; the stable refinement is the **coarsest** equitable partition of the operator, by classical Godsil. Lean: `Graphplay/Integrations/WLRefinement.lean`. This gives the Disasm worst case in polynomial time and the best case (when the operator is truly assembled by a small bundle) in a small constant.
2. **Spectral disassembly** via the coherent-algebra projection. Compute the spectral projectors of the input operator; cluster them by cell-pair signature; verify against the partition-projector algebra $\mathcal{P}(\pi)$ of Definition 3.2. When the operator is exactly cell-pair-constant, the recovered π is a faithful disassembly. Lean: `Graphplay/Dowsing/CoherentAlgebra.lean`.

9.2 The dual-rail Majorana example. The Microsoft Majorana 1 chip Hamiltonian has a 4×4 block structure at the level of pairs of Majorana zero modes. Disassembly identifies the (Q, fibers, couplings) triple as: Q is a small chiral template encoding the topological-qubit braiding pattern, fibers are trivial single-vertex sets per logical qubit, and couplings are the engineered Majorana hopping amplitudes. The resulting program runs the chiral SIGN instruction (Corollary 5.5) on the small quotient. The

applied question is which couplings produce a chiral quotient with a nontrivial speedup window; see `paper/applied_majoranal.md`.

9.3 The heavy-hex example. The IBM heavy-hex lattice has a natural 4-cell color partition: degree-2 edge vertices, degree-3 vertex bodies, and a refinement separating the two flavors of edge vertex. The resulting quotient is a 4×4 real-symmetric matrix; spectral disassembly recovers the corresponding bundle program with Q a small auxiliary graph. The flux-phasing question (under which magnetic-flux assignments does the heavy-hex lattice achieve optimal Grover search?) is then Theorem 5.3 on the 4×4 chiral quotient. Companion: `paper/applied_ibm_heavy_hex.md`.

9.4 The fragility of disassembly. A small perturbation of an operator generically breaks exact equitability; the partition collapses to the trivial single-cell partition. Robust disassembly — recovering an **approximate** program with quantitative error bounds — is open at the level of Lean mechanization, but the spectral-stability machinery of [szego2010spectral] together with the graphon limit (Theorem 6.3) supplies the natural framework. The `Toolkit/Spec.lean` stub already declares the signature of a robust disassembler with a numerical tolerance parameter.

10. Research Program

The Lean development includes thirty further modules sketching open extensions, integrations, and conjectural higher towers. The companion file `paper/research_program.typ` carries the full discussion. We list the spine in bullet form here:

- **Dowsing-rod open theorems** (`Graphplay/Dowsing/`).
 - `BundlePSTLift.lean`: PST iff quotient PST for every classical product; corner-by-corner table from Cartesian through disjunctive.
 - `ChiralBundlePST.lean`: chiral bundle PST iff signed-quotient PST; Levine signing witness `unitaryHammingChiralK4`; open questions `OpenQ1`, `OpenQ2`, `OpenQ3` on chiral product PST, K_n^σ sharpness, chiral Heawood PGST.
 - `ChiralGraphon.lean`: chiral graphons (skew-Hermitian imaginary part); `IsChiral` versus `IsReal`; equitable theory at the graphon-chiral level.
 - `CoherentAlgebra.lean`: partition-projector algebra $\mathcal{P}(\pi)$, its commutativity with Π_π , relation to coherent algebra of G .
 - `Conjecture93.lean`: the falsifiable Conjecture below with Lean apparatus and a worked example table.
 - `FilteredColimitPST.lean`: categorical PST lift along filtered colimits, Tower-5 of Theorem 6.2.
 - `FractionalRevivalNC.lean`: non-commutative fractional revival on quantum graphs; Tower-3 generalization of [chan2019fractional].
 - `HypergraphPST.lean`: k -uniform hypergraph PST and search via the bipartite incidence weighted graph.
 - `NoiseEquitable.lean`: quantitative cell-uniform preservation under generic noise.
 - `NonCommutativeCoherent.lean`: non-commutative coherent algebras for Definition 3.1 quantum graphs.
- **Cross-disciplinary integrations** (`Graphplay/Integrations/`). Each is a research-program seed pointing to an external field where the spine applies.
 - `Hodge.lean`: combinatorial Hodge decomposition; equitable partitions compatible with the cochain complex; harmonic-subspace quotient.
 - `LatticeGauge.lean`: chiral signings as discrete $U(1)$ / non-abelian connections on weighted graphs; plaquette flux.
 - `MeanFieldGames.lean`: graphon LQR control [gao2020graphon] and quantum mean-field games unified through equitable reduction.

- `OptimalTransport.lean`: graphons as transport plans, Sinkhorn alignment, quotient as a marginal-preserving lift.
- `RMT.lean`: random matrix theory meets equitable partitions; free probability on cell-uniform subspaces; free convolution for quotient spectra.
- `TensorNetworks.lean`: MERA coarse-graining as exact equitable cell maps; PEPS and holographic codes as bundles.
- `TQFT.lean`: weighted graphs as input data for $(2 + 1)$ d TQFT; modular tensor category extensions; Levin–Wen and Kitaev models.
- `WLRefinement.lean`: the Weisfeiler–Leman chain as the universal refinement of equitable partitions; algebraic and quantum WL.
- **Towers 6–7** (conjectural higher towers).
 - `Tower6.lean`: sheaves of weighted graphs over a topological base; Towers 3–5 as point-base, measure-base, sheaf-of-spectra cases; preservation of filtered colimits for the global-sections functor.
 - `Tower7.lean`: ∞ -categorical generalization; stable ∞ -categories with Hermitian endomorphisms; coherent idempotent partitions; derived Hermitian partitions; the ∞ -categorical PST lift.
- **Worked instruction-trace catalog**. Each catalog entry of Section 8 decomposes into a small instruction sequence in the combinator calculus of Section 1. For reference, the IBM heavy-hex Search entry reads `BUNDLE(Q = K_4, H = 0, B = J); PARTITION = color-classes; REFINE(marked-vertex); LIFT(Search on 4x4 quotient);` the Xie–Kay–Tamon chiral PST entry reads `BUNDLE(Q = K_2, H = path, B = J); PARTITION = 2-cell; SIGN(dual-rail phase); LIFT(PST on 2x2 chiral quotient).` A formal catalog enumeration with each program type-checked in Lean is a target for `Graphplay/Toolkit/Catalog.lean` and is the primary deliverable that turns the toolkit into a reproducible artifact.
- **Falsifiable Conjecture 9.3 (Bose–Mesner $\equiv$ chiral $\equiv$ graphon)**. A consistent sequence of finite weighted graphs (G_n, π_n) admits **both** (i) a graphon quasi-infinite limit (Theorem 6.3) and (ii) a strict chiral signing speedup on π_n (Theorem 5.3 with strict inequality on the mixing-time functional), if and only if the Bose–Mesner algebra of the family eventually coincides with the partition-projector algebra $\mathcal{P}(\pi_n)$. **Status**. The **weak** version (cross-constant signings) is believed true and supported by worked examples on K_n^σ , Hamming schemes, $K_{n,n,n,n}$, and the Heawood envelope; it is **blocked** in three confirmed both-sides-false witnesses — $K_n + P_n$, $\text{Cayley}(S_n)$ by transpositions, and any distance-regular-but-not- distance-transitive bipartite double cover. The **strong** version (arbitrary unitary signings) is conjecturally false; a candidate counterexample uses non-cross-constant phase-equitable refinements of $K_n + P_n$. Detailed analysis in `paper/conjecture93_notes.md`; Lean apparatus in `Graphplay/Dowsing/Conjecture93.lean`. The next concrete step is the Bose–Mesner-versus-distance-projector dimension table for $K_n + P_n$ at $n = 3, 4, 5$.
- **Open extensions**.
 - **Graphops**. Backhausz–Szegedy graphops generalize graphons by removing the L^2 symmetric-kernel assumption. The quotient spine survives when the operator commutes with Π ; Theorem 6.3 should extend to graphops under mild compactness.
 - **Surface topology**. The Heawood map-color completion is the bundle with $Q = K_{H(g)}$; the surface algebraic-connectivity ceiling of [freitas2001heawood] is a Tower-2 spectral budget on Q .
 - **Open systems and time dependence**. Bundle-and-quotient survives unitary perturbations whose generator is equitable, and cell-uniform-symmetric noise (`Toolkit/Noise.lean`). Lindbladian extensions and general time-dependent schedules are not yet covered beyond the cell-uniform invariance lemma.

- **Applied case studies.** `paper/applied_majorana1.md` (Microsoft Majorana 1 topological-qubit fabric) and `paper/applied_ibm_heavy_hex.md` (IBM heavy-hex lattice) are sibling-agent companions written against the Tower-2 / Tower-3 spine.

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